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Raritone Summer Internship 2026

AI/ML Team – Pose Detection Accuracy Analysis Report

Submitted By: Member 2

Objective

The objective of this task is to study, implement, and analyze pose detection models used in AI-based Virtual Try-On systems. Pose detection helps identify human body landmarks and plays a crucial role in body measurement extraction, clothing alignment, and virtual garment fitting.

The primary goal of this experiment was to evaluate pose detection accuracy under different conditions such as front view, side view, various body poses, and movement scenarios.

Tools Used

- Python
- MediaPipe Pose
- OpenCV
- Visual Studio Code

Introduction

Pose detection is a computer vision technique that identifies key points on the human body, including shoulders, elbows, wrists, hips, knees, and ankles. These key points are called landmarks and are used to understand human posture and body structure.

In Virtual Try-On systems, pose detection helps align garments according to the user's body position, improving realism and fitting accuracy.

MediaPipe Pose was selected for implementation because it provides real-time body landmark detection with high speed and accuracy.

Implementation

A real-time pose detection system was developed using Python, OpenCV, and MediaPipe Pose.

The webcam was used as the input source. Frames captured from the webcam were processed using MediaPipe's pose estimation model. The detected landmarks and skeletal connections were displayed on the screen in real time.

The application successfully detected and tracked 33 body landmarks while maintaining smooth real-time performance.

Experimental Procedure

The following tests were conducted:

Test 1: Front View Detection

The user stood directly facing the camera. Landmark positions were observed and analyzed.

Test 2: Side View Detection

The user turned sideways to evaluate landmark detection when body parts were partially hidden.

Test 3: Raised Hands Pose

Both hands were raised to test shoulder, elbow, and wrist tracking accuracy.

Test 4: Dynamic Movement

The user performed simple movements to evaluate tracking stability.

Test 5: Different Lighting Conditions

The pose detector was tested under normal and reduced lighting conditions.

Results and Observations

Front View

The model accurately detected all major body landmarks. Skeleton tracking remained stable and consistent.

Observation:

Excellent landmark visibility and high confidence scores.

Side View

Most landmarks were successfully detected even when some body parts became partially hidden.

Observation:

Minor reduction in confidence for occluded joints.

Raised Hands Pose

The pose detector successfully tracked arm movement and accurately identified shoulder, elbow, and wrist landmarks.

Observation:

Very stable upper-body landmark detection.

Dynamic Movement

The model maintained continuous landmark tracking during movement.

Observation:

Smooth real-time tracking with minimal lag.

Lighting Conditions

Under normal lighting, landmark detection was highly accurate. Under lower lighting conditions, slight fluctuations in landmark confidence were observed.

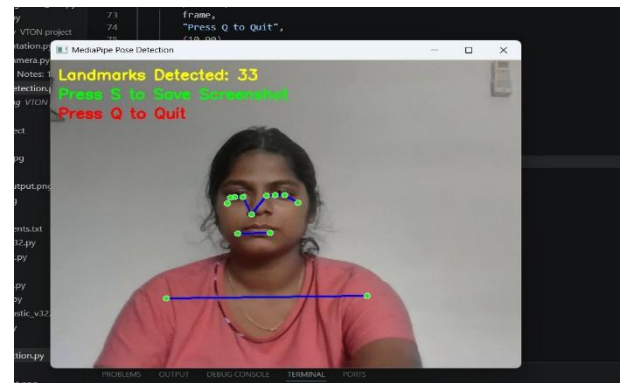
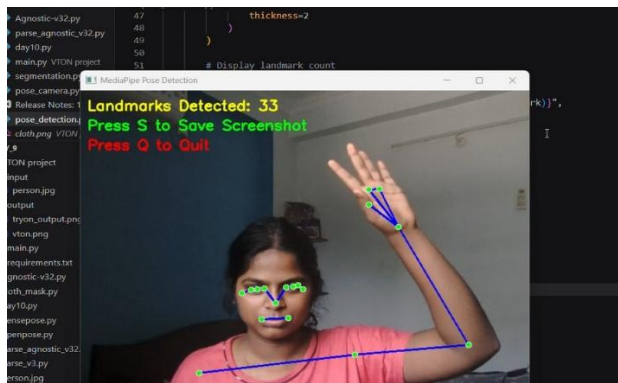
Observation:

Performance decreases slightly as lighting quality reduces.

```
pose_detection.py > ...
1  import cv2
2  import mediapipe as mp
3  import time
4  mp_pose = mp.solutions.pose
5  mp_drawing = mp.solutions.drawing_utils
6  pose = mp_pose.Pose(
7      static_image_mode=False,
8      model_complexity=1,
9      min_detection_confidence=0.5,
10     min_tracking_confidence=0.5
11 )
12 cap = cv2.VideoCapture(0)
13 screenshot_count = 1
14 while True:
15     success, frame = cap.read()
16     if not success:
17         break
18     frame = cv2.flip(frame, 1)
19     rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
20     results = pose.process(rgb)
21     if results.pose_landmarks:
22         mp_drawing.draw_landmarks(
23             frame,
24             results.pose_landmarks,
25             mp_pose.POSE_CONNECTIONS,
26             mp_drawing.DrawingSpec(
27                 color=(0,255,0),
28                 thickness=2,
29                 circle_radius=2
30             ),
31             mp_drawing.DrawingSpec(
32                 color=(255,0,0),
33                 thickness=2
```

pose_detection.py > ...

```
35         )
36         cv2.putText(
37             frame,
38             f"Landmarks Detected: {len(results.pose_landmarks.landmark)}",
39             (10,30),
40             cv2.FONT_HERSHEY_SIMPLEX,
41             0.7,
42             (0,255,255),
43             2
44         )
45     cv2.putText(
46         frame,
47         "Press S to Save Screenshot",
48         (10,60),
49         cv2.FONT_HERSHEY_SIMPLEX,
50         0.7,
51         (0,255,0),
52         2
53     )
54     cv2.putText(
55         frame,
56         "Press Q to Quit",
57         (10,90),
58         cv2.FONT_HERSHEY_SIMPLEX,
59         0.7,
60         (0,0,255),
61         2
62     )
63     cv2.imshow("MediaPipe Pose Detection", frame)
64     key = cv2.waitKey(1)
65     if key == ord('s'):
66         filename = f"pose_output_{screenshot_count}.png"
67         cv2.imwrite(filename, frame)
68         print(f"Saved: {filename}")
69         screenshot_count += 1
70     elif key == ord('q'):
71         break
72 cap.release()
73 cv2.destroyAllWindows()
```



Accuracy Analysis

The MediaPipe Pose model demonstrated strong performance across all test conditions.

Key observations include:

- Accurate detection of 33 body landmarks.
- Stable tracking during movement.
- Fast real-time processing.
- Reliable performance under normal lighting.
- Slight reduction in confidence during low-light conditions and side poses.

Overall, the model achieved high landmark detection accuracy suitable for Virtual Try-On applications.

Advantages of MediaPipe Pose

- Real-time performance
- Lightweight architecture
- Easy implementation
- Accurate body landmark detection
- Low hardware requirements
- Suitable for webcam-based applications

Limitations

- Reduced accuracy under poor lighting.
- Performance may decrease when body parts are heavily occluded.
- Extreme body poses can affect landmark confidence.

Applications in Virtual Try-On

Pose detection is an essential component of Virtual Try-On systems because it enables:

- Body landmark identification
- Body measurement estimation
- Clothing alignment
- Garment fitting
- User posture understanding

Accurate pose estimation directly improves the realism and fitting quality of virtual clothing.

Conclusion

A real-time pose detection system was successfully implemented using MediaPipe Pose and OpenCV. The system accurately detected and tracked 33 body landmarks under different body poses and movement conditions.

Experimental results showed that MediaPipe Pose provides an effective balance between speed, accuracy, and ease of implementation. The model is highly suitable for Virtual Try-On applications where real-time body landmark detection is required.

The experiment confirmed that pose detection serves as a foundational step in building reliable AI-based Virtual Try-On systems.